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Report

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Contents

Introduction **2**

1 BB84 Protocol, Quantum Key Distribution **3**

1.1 Working Principle . . . 3

1.2 Presence of Eve . . . 3

1.3 Advantage of BB84 . . . 4

2 SPAD Detection Efficiency and Characterization **5**

2.1 Objective of the Experiment . . . 5

2.2 Operating Principles of the SPAD . . . 5

2.3 Detection Efficiency and Substitution Method . . . 6

3 Backflash Propagation and Detection Security **8**

3.1 Backflash Emission and Security Risks . . . 8

3.2 Experimental Setup: SP-OTDR . . . 8

3.3 Information Leakage Estimation . . . 9

Introduction

In the field of communications, the need to protect information from unauthorized external accesses has always been a priority over the centuries. The science that studies this is cryptography, which has the objective to study algorithms (cypher) able to encode the original message combining it with an additional information (the key) generating therefore a cyphertext, that can be decoded only by who possesses exactly the correct key. There are two main classes of cryptographic algorithms:

- Asymmetric Cryptographic algorithms: They are based on a couple of different keys. The sender (Alice, A) encodes the message using a public key, while the receiver (Bob, B) uses its own secret private key to decode it.
- Symmetric Cryptographic algorithms: There is a unique private key shared between Alice and Bob, used for both encoding and decoding and that needs to be exchanged through a secure channel.

Nowadays, one of the most used classical asymmetric algorithms is the RSA. It is based on the prime numbers properties and modular algebra. The problem is that the RSA security is purely computational, meaning it is based on the high complexity for a classical computer to factorize very big prime numbers. However, the possible advent of quantum computers and the Shor's algorithm can allow to solve this mathematical problem in much shorter times, making the RSA vulnerable.

To overcome this problem, the solution is therefore to use an algorithm that is theoretically impossible to decrypt, that is the One-Time Pad (OTP), a symmetric algorithm based on the XOR logical operator between binary numbers. The algorithm consists in the following steps:

1. A translates the message M in binary and applies the XOR with a random key K of the same length of M ($C = M \oplus K$).
2. B decodes the message applying the same operation on the cyphertext ($M = C \oplus K$).

Although the one-time pad is theoretically unbreakable, its security depends on the key being truly random, kept secret, and at least as long as the message. Consequently, if an attacker were able to intercept the secure channel used to exchange the key through a "man in the middle" attack, for example, they could obtain the key and completely compromise the message. Quantum mechanics was introduced to address this key exchange problem through protocols such as BB84, a quantum key distribution (QKD) protocol.

Chapter 1

BB84 Protocol, Quantum Key Distribution

BB84 is a quantum key distribution protocol that allows two parties, Alice (A) and Bob (B), to generate a shared secret key K over an insecure channel, while allowing them to detect disturbances in the quantum communication channel by monitoring the Quantum Bit Error Rate (QBER), as discussed later. It exploits the principles of quantum mechanics, in particular the no cloning theorem, which states that it is impossible to create an identical copy of an arbitrary unknown quantum state, and the non-orthogonality of states belonging to different polarization bases. Since these states cannot be perfectly distinguished, an eavesdropper (Eve) attempting to measure them may introduce detectable disturbances.

1.1 Working Principle

The protocol consists of the following steps:

1. Alice randomly generates a sequence of bits and for each bit randomly chooses one of the two polarization bases: the linear basis $Z = \{|H\rangle, |V\rangle\}$ or the diagonal basis $X = \{|+\rangle, |-\rangle\}$.

Table 1.1: Encoding of classical bits into polarization states in the BB84 protocol.

Bit	**Rectilinear basis (Z)**	**Diagonal basis (X)**
0	$\|H\rangle$ (0)	$\|+\rangle$ $(\pi/4)$
1	$\|V\rangle$ $(\pi/2)$	$\|-\rangle$ $(-\pi/4)$

2. B doesn't know the polarization state of the incoming photon, so he independently and randomly chooses a measurement basis (Z or X) to read each single qubit. To do this, B uses his own polarization rotator followed by a PBS (Polarizing Beam Splitter) and two detectors to measure the photon's state.
3. If B happens to choose the same basis used by A, he measures the photon correctly with 100% probability; however, when Bob chooses a different basis, the measurement outcome becomes completely random, with a 50% probability for each possible bit value.
4. Once all the photons are sent, A and B use a classical public channel to compare the sequences of the bases they used for each photon and they remove all the bits where their bases didn't coincide. This step is called "sifting".
5. At this point, they publicly compare a random sample of their sifted bits to estimate the QBER. Even when their bases coincide, experimental imperfections may introduce errors.

1.2 Presence of Eve

Although BB84 does not prevent Eve from intercepting the transmitted photons, an interception attempt may introduce detectable errors, as stated before.

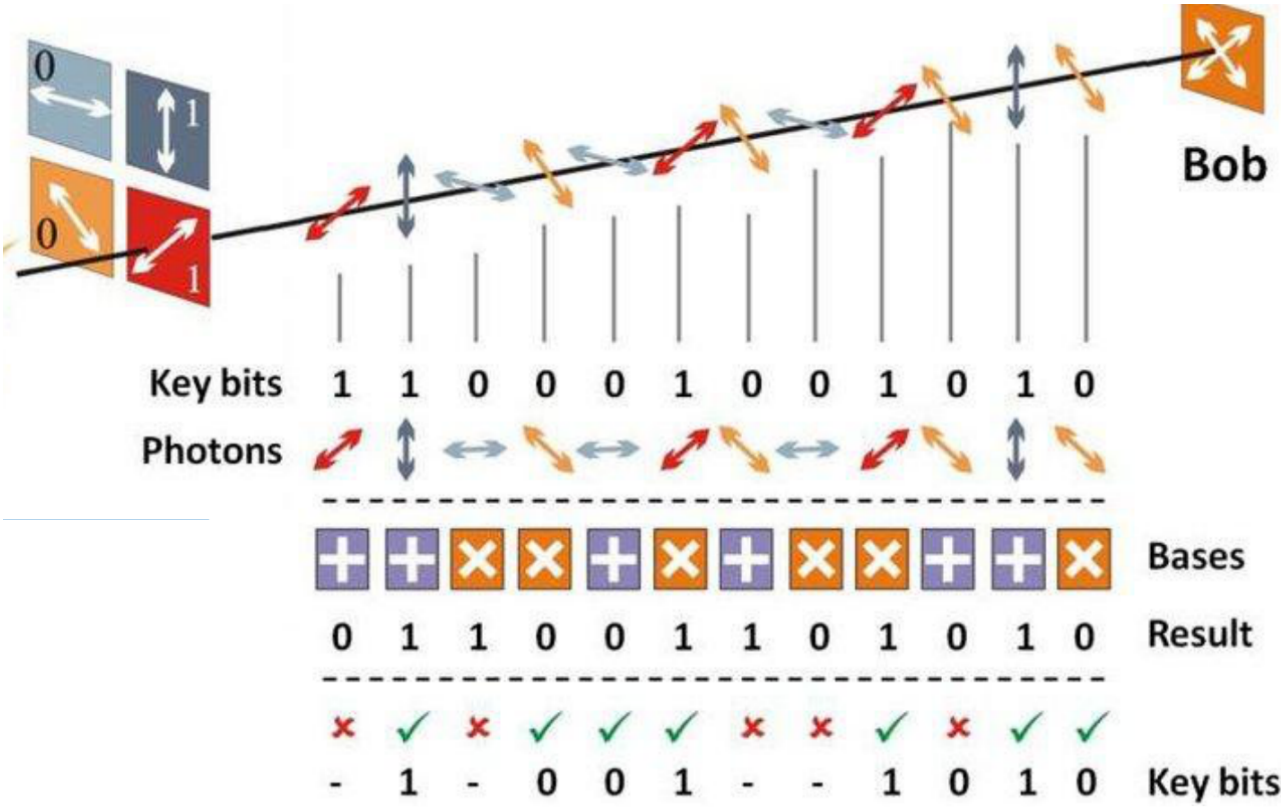


Figure 1.1: Polarization states, measurement bases, and sifting in the BB84 protocol.

A simple example is the intercept-resend attack, in which Eve measures each intercepted photon in a randomly chosen polarization basis and then prepares a new photon in the measured state to send to Bob. Since Eve does not know Alice's preparation basis, she has a 50% chance of choosing the wrong one. In this case, even when Bob measures in the correct basis (Alice's basis), he has a probability of 1/2 of obtaining the wrong bit.

Therefore, assuming an ideal scenario with no experimental imperfections, the Quantum Bit Error Rate among the sifted bits is

$$\mathrm{QBER} = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4} = 25\%.$$

1.3 Advantage of BB84

By publicly comparing a randomly selected subset of their sifted bits, Alice and Bob can estimate the Quantum Bit Error Rate (QBER), which quantifies the discrepancies between their measurement outcomes.

A high QBER may indicate the presence of an eavesdropper, although errors can also arise from experimental imperfections and detector noise. For instance, an ideal intercept-resend attack introduces a QBER of 25%, but this value does not represent a universal security threshold. If the estimated QBER is sufficiently low to allow secure key extraction, Alice and Bob can apply error correction and privacy amplification to the remaining bits. The resulting secret key can then be used for classical encryption, for example through the One-Time Pad (OTP).

However, the security of a practical QKD system also depends on the physical implementation of its components, since imperfections in single-photon detectors may introduce additional vulnerabilities.

Chapter 2

SPAD Detection Efficiency and Characterization

2.1 Objective of the Experiment

Single-photon detectors are essential components of a QKD system. However, unlike ideal detectors, real devices exhibit finite detection efficiency and limitations that affect their performance and security. In this experiment, we will focus on characterizing the performance of a Single-Photon Avalanche Diode (SPAD) by measuring its detection efficiency using the substitution method.

2.2 Operating Principles of the SPAD

To implement the BB84 protocol, a detector capable of registering individual photons is required. A commonly used device is the SPAD (Single Photon Avalanche Diode), a semiconductor photodiode reverse-biased above its breakdown voltage. When the photon is absorbed in the active region, it can create an electron-hole pair. The resulting charge carriers are accelerated by the strong electric field in the multiplication region. Here, the electric field is at its maximum and the carriers are accelerated so much that they trigger a self-sustaining impact ionization avalanche. This translates into a macroscopic output current, making us able to detect the single photon.

- **Geiger Mode and Quenching:** The SPAD works in Geiger Mode. Since the arrival of a single photon can trigger a macroscopic avalanche current, any subsequent photon arriving during the avalanche has no additional measurable effect, meaning the device cannot count the exact number of simultaneous photons. For this reason, a quenching circuit (often a passive circuit using a resistor) is necessary to stop the avalanche by decreasing the voltage below the breakdown threshold, and then restore the overvoltage. During quenching and resetting, the detector is temporarily unable to register new photons. This dead time sets an upper limit on the achievable count rate.
- **Afterpulsing and Hold-off Time:** During the avalanche process, some carriers can remain trapped by defects in the SPAD material and can be released with a certain delay, triggering a second avalanche (afterpulsing). To reduce this effect, a period of forced inactivity is imposed after each detection, called the *hold-off time*, allowing trapped carriers to be released before the detector is reactivated, thereby reducing the probability of afterpulsing.
- **Dark Count Rate (DCR):** Thermal excitation or material defects can generate charge carriers that trigger an avalanche even if no photon has arrived, resulting in a false detection (dark count).
- **Timing Jitter:** It represents the statistical fluctuation of the time delay between the actual photon arrival and the generation of the macroscopic output pulse.

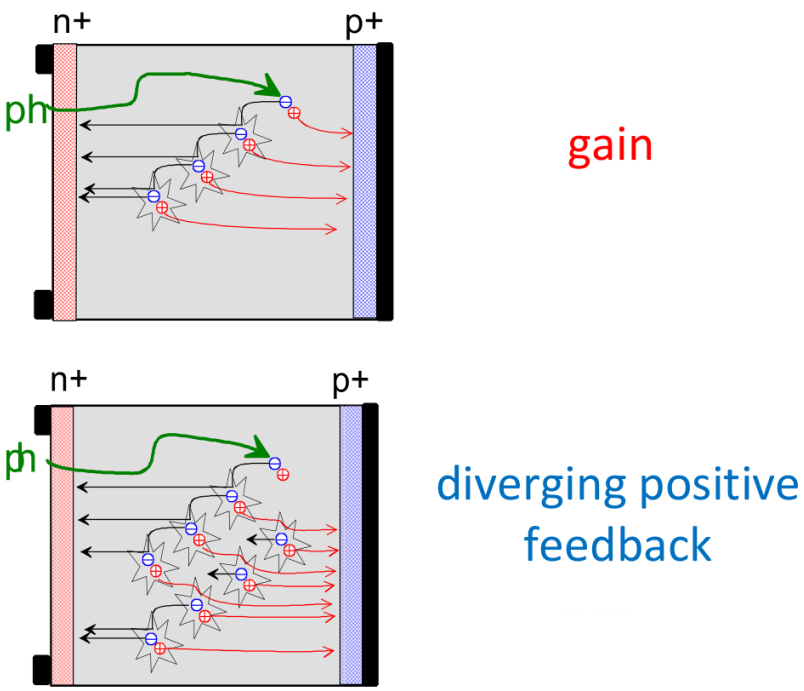


Figure 2.1: Comparison of avalanche multiplication in linear APD operation (top) and Geiger-mode SPAD operation (bottom).

2.3 Detection Efficiency and Substitution Method

The SPAD Detection Efficiency is the probability that an incident photon successfully yields a count. It can be calculated as the ratio between the detected signal count rate and the incident photon flux.

To measure this, the substitution method setup can be used. A continuous laser beam is divided into two paths: one path goes into a reference Optical Power Meter to measure the optical power (in watts), while the other path goes through a Variable Optical Attenuator (VOA) that reduces the light to a low photon-flux regime before reaching the SPAD (Detector Under Test).

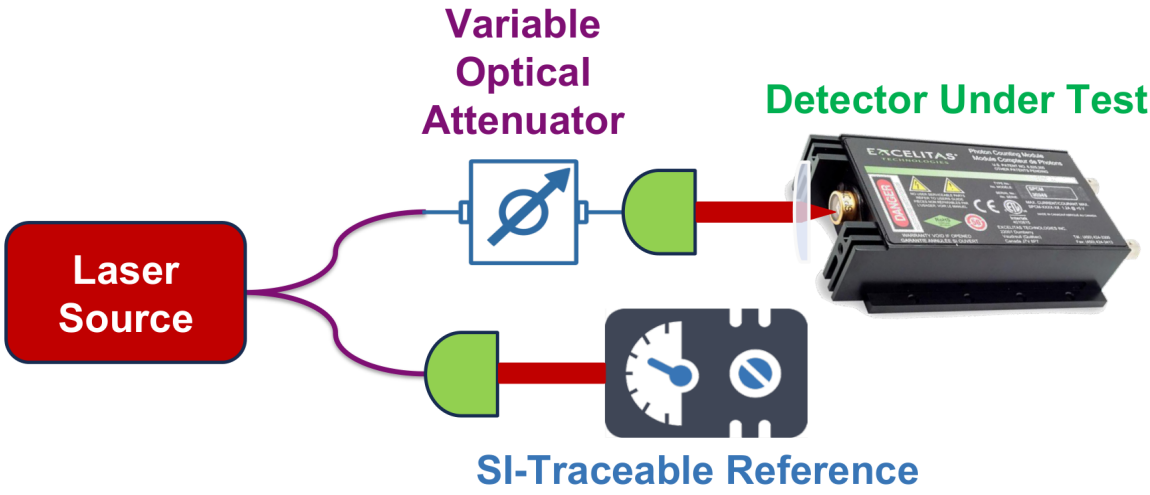


Figure 2.2: Experimental setup for SPAD detection efficiency characterization using the substitution method.

The detection efficiency can be calculated as:

$$\eta\left(\lambda, \frac{N}{t}\right) = \frac{hc\,(N - N_{\text{env}})}{\lambda t \tau C\,(P - P_{\text{env}})} \tag{2.1}$$

where:

- $\frac{N-N_{\text{env}}}{t}$ is the net signal count rate, obtained by subtracting the background counts N_{env} from the total counts N registered during the measurement time t.
- $\frac{\lambda(P-P_{\text{env}})}{hc}$ is the photon flux corresponding to the background-corrected reference optical power, where $E = hc/\lambda$ is the energy of a single photon.
- τ is the transmittance of the optical attenuator.
- C is the calibration factor relating the reference power measurement to the optical power entering the attenuated path.

However, the apparent detection efficiency decreases as the incident photon flux increases, since more photons arrive during the detector dead time. For this reason, the zero-flux detection efficiency, $\eta_0(\lambda)$, is defined as the limiting efficiency when the incident photon flux tends to zero.

Under non-paralyzable dead-time considerations and neglecting background contributions, the following relation can be used:

$$\eta\left(\lambda, \frac{N}{t}\right) = \eta_0(\lambda)\left(1 - \frac{D_t N}{t}\right) \tag{2.2}$$

where D_t represents the detector dead time.

By measuring the detection efficiency at different count rates, $\eta_0(\lambda)$ can be estimated from the intercept of a linear fit in the low-count-rate regime.

Chapter 3

Backflash Propagation and Detection Security

3.1 Backflash Emission and Security Risks

Although BB84 can reveal certain eavesdropping attacks through an increase in the QBER, real SPAD detectors may introduce additional security vulnerabilities.

During the avalanche process, the SPAD can emit photons that propagate backwards into the optical fiber. These are called *backflash photons*. Eve could potentially detect these photons and deduce which of Bob's detectors registered a click, without necessarily introducing additional errors in the quantum communication channel.

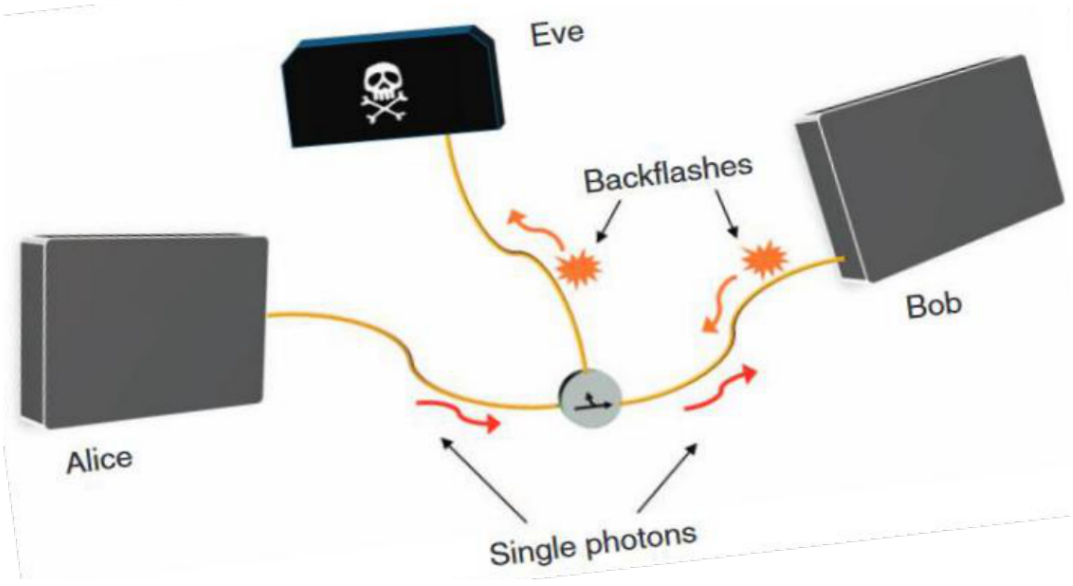


Figure 3.1: Schematic representation of a potential backflash side-channel attack in a QKD system.

3.2 Experimental Setup: SP-OTDR

The backflash emission is characterized using a Single-Photon Optical Time-Domain Reflectometry (SP-OTDR) setup as shown in Figure 3.2.

A pulsed laser source is attenuated to the single-photon level and directed towards the Detector Under Test (DUT).

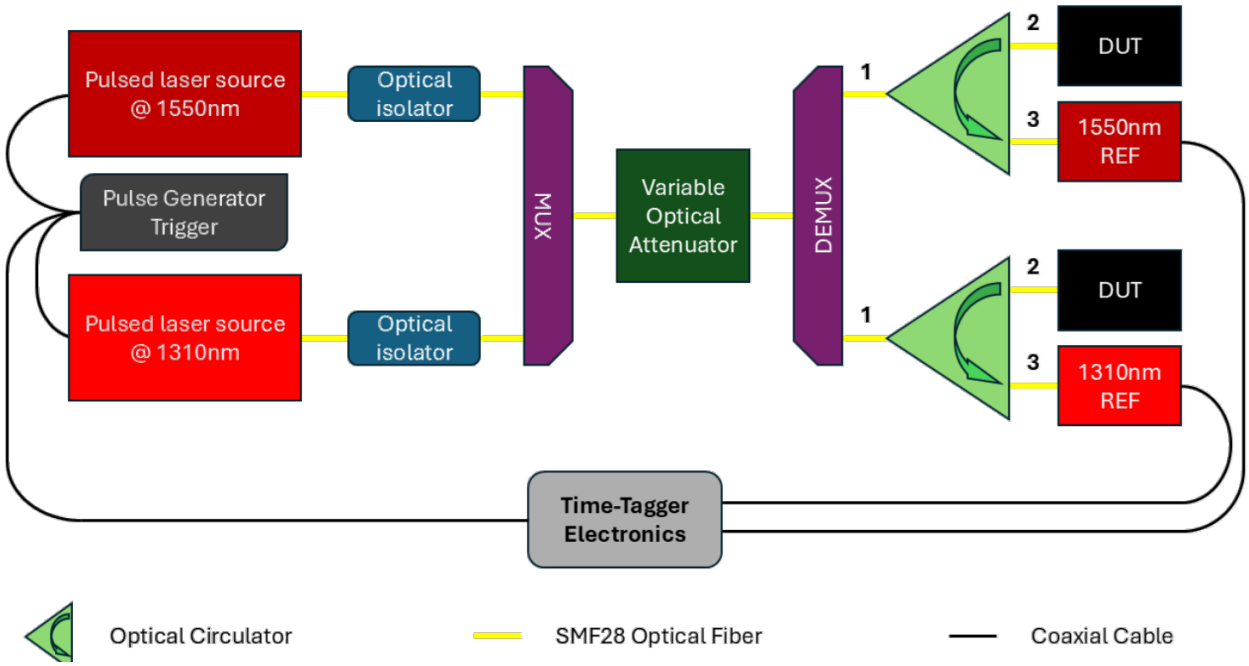


Figure 3.2: Experimental SP-OTDR setup for backflash characterization using pulsed laser sources, optical circulators, reference SPADs and time-tagging electronics.

When a photon triggers an avalanche, the DUT may emit backflash photons that propagate backwards through the optical fiber. An optical circulator redirects the backward propagating light towards a reference InGaAs/InP SPAD detector. By recording the detection times of both detectors, temporal correlations can be analyzed to identify the backflash signal and distinguish it from background events and optical reflections.

3.3 Information Leakage Estimation

The backflash emission can be quantified by analyzing the temporal correlations between the DUT and the reference detector.

After subtracting the background contribution, the information leakage parameter can be estimated as

$$P_L = \frac{N_{\text{bf}}}{N_{\text{DUT}} \eta_{\text{CH}} \eta_{\text{OTDR}}} \tag{3.1}$$

where:

- N_{bf} is the net number of detected backflash photons.
- N_{DUT} is the total number of valid detections registered by the DUT.
- η_{CH} accounts for the optical transmission losses between the DUT and the reference detector.
- η_{OTDR} is the detection efficiency of the reference SPAD.